

DSA STUDY BLUEPRINT SERIES

78. Binary Trees in Data Structures Tree Traversal

Hierarchical Tree structures and Traversals

Section 09 • C++ Core Data Structures & Algorithms

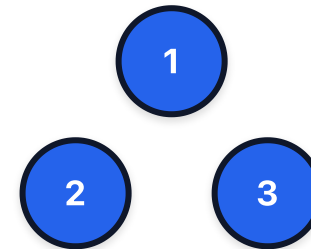
Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Traversals:** Depth Exploration: Preorder (N-L-R), Inorder (L-N-R), Postorder (L-R-N).
- **BFS level-order:** Traverses tree nodes level-by-level using a queue.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
struct TreeNode {  
    int val; TreeNode *left, *right;  
    TreeNode(int x) : val(x), left(nullptr), right(nullptr) {}  
};  
  
void inorder(TreeNode* root) {  
    if (!root) return;  
    inorder(root->left);  
    cout << root->val << " ";  
    inorder(root->right);  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Traversals (All nodes)	$O(N)$ time, $O(H)$ space

Key Practices & Gotchas

- **Important:** Recursion stack space equals tree height ($O(H)$). Worst case is $O(N)$ for skewed trees.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace recursive Preorder Traversal visits (Node-Left-Right) for tree `1 (left=2, right=3)`:

Call Stack Level	Active Node Visited	Expected Left Traversal	Expected Right Traversal
1	Root (1)	Recurse on Left child (2)	Recurse on Right child (3)
2	Leaf (2)	Base case (nullptr)	Base case (nullptr)

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Iterative traversals:** Standard DFS traversals can run iteratively by simulating recursion calls with Stacks.
- **Level Order BFS:** Level tracking searches use Queues to visit node layers.
- **Related LeetCode Problems:** #102 Binary Tree Level Order Traversal, #94 Binary Tree Inorder Traversal, #236 Lowest Common Ancestor.